

Localization of Single Vibrotactile Cues in Forearm-Mounted Multi-Tactor Displays

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I. INTRODUCTION

Wearable vibrotactile displays have gained attention as an effective means of conveying information through the sense of touch, particularly when visual or auditory channels are unavailable [1]. The forearm provides a practical site for wearable displays. Advances in compact actuators have made it feasible to mount multiple tactors on this region.

A key question is whether users can accurately localize individual vibrotactile stimuli delivered through multi-tactor arrays on the forearm. Previous studies have shown that localization performance is affected by the number and placement of tactors [2], and that performance degrades as more tactors are simultaneously activated [1]. Huang et al. further examined how intensity and frequency influence spatial acuity, though without systematically comparing different anatomical regions [3].

However, there remains no clear guideline for how tactors should be spatially arranged to maximize localization accuracy. Prior work has largely focused on arbitrary configurations, leaving the effects of structured spatial layouts underexplored. This limits their utility in applications requiring precise tactile cues, such as notifications or simple communication.

In this study, we evaluated how accurately users can localize vibrotactile stimuli when different numbers of tactors (four, six, or eight) are placed on specific parts of the forearm—either near the wrist (distal) or near the elbow (proximal). The distal and proximal regions were selected to promote perceptual discrimination. Our goal is to examine how the number and configuration of tactors affect recognition accuracy, and to derive practical implications for designing forearm-mounted vibrotactile interfaces.

II. METHODS AND USER STUDY

A. Apparatus

To deliver vibrotactile stimuli with strong perceived intensity, we used eccentric rotating mass (ERM) motors instead of linear resonant actuators (LRAs). The motors (Seeed Technology, 10.00 mm diameter) were driven by an Arduino Due, which supports simultaneous control of multiple PWM channels. Tactors were mounted on a velcro band, allowing their positions to be adjusted based on the user's forearm

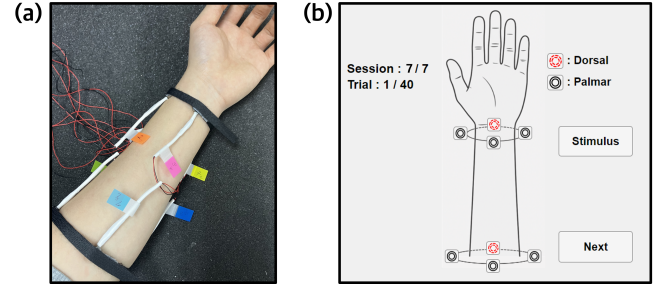


Fig. 1. (a) Experimental setup showing the placement of tactors on the distal and proximal regions of the forearm using a velcro band. (b) User interface used for stimulus delivery and response input during the experiment.

size. As shown in Fig. 1(a), the band was worn either on the distal or proximal region of the forearm to evaluate localization performance at two anatomically distinct sites.

B. Experimental Conditions

Seven tactor configurations were tested across seven sessions, as illustrated in Fig. 2. In Sessions 1 and 2, four tactors were placed along either the palmar–dorsal or medial–lateral axis on a single region of the forearm. Sessions 3 and 4 used six tactors, with three on the distal region and three on the proximal region. Each region included tactors on the medial, lateral, and palmar sides in Session 3, and on the medial, lateral, and dorsal sides in Session 4. In Sessions 5 and 6, the same number of tactors were used, but with two tactors placed on the palmar side and one on the dorsal side (Session 5), or vice versa (Session 6), to vary inter-tactor spacing. Session 7 used eight tactors placed on the dorsal, palmar, medial, and lateral sides of both distal and proximal regions, representing the most spatially dense configuration. Tactors were labeled anatomically (e.g., D1–D3 for distal, P1–P5 for proximal), as shown in Fig. 2(b)

C. Procedure

Ten participants (six male and four female, mean age: 25.1 ± 2.4 years) took part in the experiment. All procedures were approved by the institutional review board (IRB protocol: PIRB-2023-E025).

Before each session, participants were shown the layout of the active tactors through the user interface (UI), as shown in Fig. 1(b). Upon pressing the *Stimulus* button, a single vibrotactile stimulus lasting two seconds was delivered from one of the active tactors selected at random. Each tactor was activated five times in a randomized order per session. After

Research supported by Support Program for xxx funded by the xxx (Project Number: xxx).

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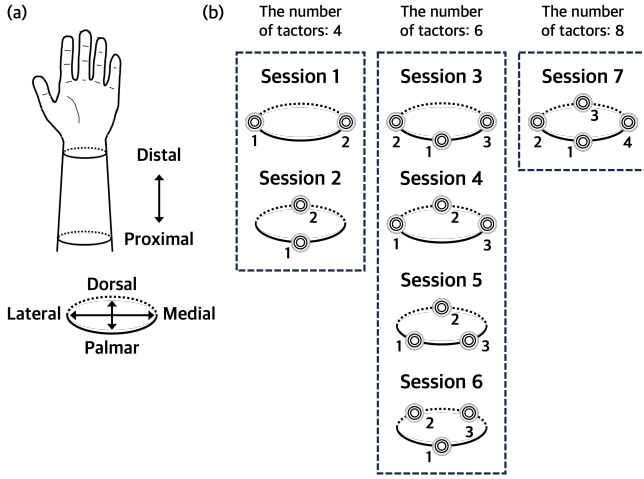


Fig. 2. (a) Anatomical reference for the distal and proximal regions of the forearm used in the experiment. (b) Tactor configurations for each session. The cross-sectional diagrams illustrate the spatial configuration of tactors, and the numbers indicate the label assigned to each tactor during the corresponding session.

each stimulus, participants identified the perceived location by selecting one of the predefined positions in the UI and proceeded to the next trial by pressing the *Next* button.

III. RESULTS AND DISCUSSION

Localization performance was evaluated across seven actuator configurations with varying numbers and spatial distributions of tactors. Recognition accuracy exceeded 95% in all sessions, indicating reliable discrimination of spatial tactile cues on the forearm.

Figure 3 presents the confusion matrices for each session. Sessions 1 and 2, which used four tactors, achieved 99.0% and 100.0% accuracy, respectively. These results suggest that localization performance remains consistently high with four tactors, regardless of configuration.

Sessions 3 to 6 used six tactors and showed robust accuracy, ranging from 97.7% to 99.3%. Among these, a trend was observed in which configurations that avoided the medial and lateral sides yielded fewer misclassifications, suggesting that certain configurations may reduce localization confusion.

Session 7, which used eight tactors, resulted in 95.5% accuracy. Although performance slightly decreased with increased tactor count, accuracy remained high enough to support the feasibility of denser configurations.

Overall, these results indicate that up to six tactors can be reliably localized across configurations, and even eight tactors can be used effectively without substantial degradation in performance.

IV. CONCLUSION AND FUTURE WORK

All configurations tested in this study yielded high localization accuracy, with recognition rates exceeding 95% across conditions. These results demonstrate the feasibility of forearm-mounted vibrotactile displays using up to eight tactors.

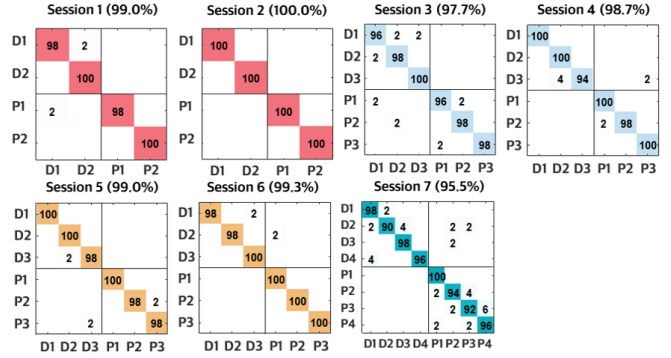


Fig. 3. Confusion matrices for each session. The x- and y-axis labels indicate tactor locations using anatomical codes: 'D' and 'P' refer to the distal and proximal regions of the forearm, respectively, and the numbers represent the tactor labels assigned in each configuration.

Future work will first focus on evaluating localization performance with more than eight tactors to assess the scalability of forearm-mounted arrays. We will then extend the evaluation to cover the entire forearm, aiming to develop spatial design guidelines that maximize localization accuracy across different anatomical regions.

Subsequent studies will investigate how temporal factors, including stimulus intensity, duration, and motion dynamics, influence localization performance. Based on these findings, we plan to establish optimized temporal parameters for tactile stimulus design.

Ultimately, this line of research seeks to develop structured tactile languages capable of conveying complex information through spatial and temporal patterns on the forearm.

REFERENCES

- [1] I. Oakley, Y. Kim, J. Lee, and J. Ryu, "Determining the feasibility of forearm mounted vibrotactile displays," in *2006 14th symposium on haptic interfaces for virtual environment and teleoperator systems*. IEEE, 2006, pp. 27–34.
- [2] D. Wang, C. Peng, N. Afzal, W. Li, D. Wu, and Y. Zhang, "Localization performance of multiple vibrotactile cues on both arms," *IEEE Transactions on Haptics*, vol. 11, no. 1, pp. 97–106, 2017.
- [3] B. Huang, P. H. Dietz, and D. Wigdor, "Investigating the effects of intensity and frequency on vibrotactile spatial acuity," *IEEE Transactions on Haptics*, vol. 17, no. 3, pp. 405–416, 2024.